Beyond Aging Clocks

Measuring biological function over time

Ryan J. McCall, PhD · Longevity Biotech Fellowship

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The Problem with Current Aging Clocks

- An aging clock gives you a static number
- Moving that number with a treatment has never been shown to make an animal live longer
- Regulators do not currently accept clocks as clinical trial endpoints
- Trials count actual disease events instead
- A clock tracks the slowest-recovering combination of measurements in the body, which takes about a lifetime, while recovery itself happens in weeks.
- A number measured on the slow scale cannot give you a rate on the fast one.

State vs. Function

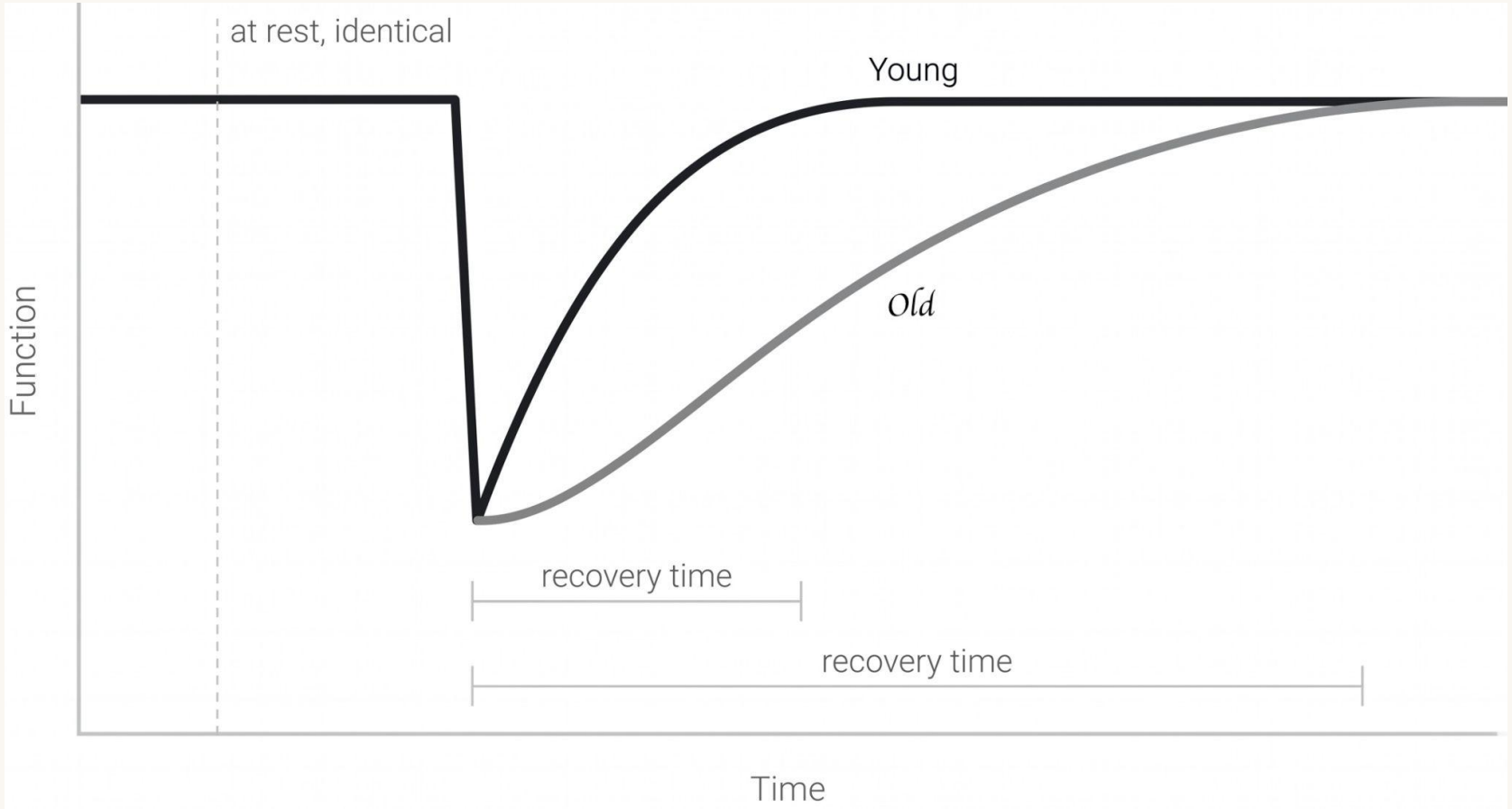
- **State:** a snapshot of molecules at one moment
- **Function:** how fast the system recovers after a perturbation

- Aging is a decline in function
- Clocks measure state

- Almost every dataset we have is state
- Even perturbation data: the moment after the dose, not the recovery

Recovery Rate Already Predicts Mortality

- In large human cohorts, variables that return slowly to baseline carry higher mortality risk
- Every one of those recovery rates is inferred passively, from years of watching
- Nobody delivers a perturbation, at a time they choose, and measures the recovery



Two Ways to Measure Recovery

Perturb and time. Fast, but tissue in a dish never dies.

Watch and wait. Death is real, but one animal takes eight months.

The hypothesis. Both measure the same recovery.

The plan. Tissue finds the measurement. Animals prove it predicts death.

What Public Data Can and Cannot Answer

Brain tissue. Two public datasets, both from 2022.

- **They can test whether my measure is repeatable.**
- **They cannot test whether a drug changes it.**
  No well got the liquid without the drug, recorded over the same hours.

Killifish. Too few deaths. Noise beats every score I measured.

Recovery is a change over time. So is drift.
Without that control, you cannot tell them apart.

Start with the Tissue

What we build: recovery profiles across donors, ages and compounds.

Is it repeatable?
Is it the tissue, or my equipment?
Is it better than what we already have?

Twenty wells. Two weeks. All three answered.
If a gate fails, I rebuild the protocol. The question stays.

If it Passes, the Organism Track

Only if the tissue measure clears all three hurdles.

An open challenge. Predict how much life an animal has left.
Killifish first, then mouse.
Rules published before anyone enters.

The only place a readout can be checked against death.

$1.8 billion was just committed to measuring how cells respond. What it measures is still being decided.

Where I Could Use Help

- **Recordings you already have.** Anything followed to the end of life.
- **Twenty minutes, if you run an electrode array.**
- **Three or four people to fix the scoring rules.**

Conclusion

- Aging is strongly associated with slower recovery. Nobody has built the protocol to measure it, at either scale.
- **If you had to pick one readout to validate against death, what would you pick?**

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